Q1 - Single Correct

If p is the prime number and $p \neq 2$, then $\left[(2+\sqrt{5})^p\right]-(2)^{p+1}$ is always divisible by (where, $[x]$ denotes greatest integer function less than or equals to x)

(1) $p+1$

(2) p

(3) $2p$

(4) $4p$

Q2 - Single Correct

The number 7^{1995}, when divided by 100 leaves the remainder

(1) 43

(2) 45

(3) 50

(4) None of these

Q3 - Single Correct

The coefficient of x^9 in the polynomial given by , $(x+1)(x+2)\cdots(x+10)+(x+2)$ $(x+3)\cdots(x+11)+\cdots+(x+11)(x+12)\cdots(x+20)$, is

(1) 5511

(2) 5151

(3) 1515

(4) 1155

Q4 - Single Correct

In the expansion of $(1+x)^n(1+y)^n(1+z)^n$, the sum of the coefficients of the terms of degree r, is

(1) ${}^{n^3}\mathrm{C}_r$

(2) ${}^{n}C_{r^3}$

(3) ${}^{3n}\mathrm{C}_r$

(4) $3 \cdot {}^{2n}C_r$

Q5 - Single Correct

The value of $\sum_{r=0}^{n}(-1)^r\left(\frac{{}^nC_r}{{}^{r+3}C_r}\right)$, is

(1) $\frac{n(n+1)}{2}$

(2) $\frac{n+3}{3}$

(3) $\frac{3}{n+3}$

(4) $\frac{n+2}{2}$

Q6 - Single Correct

The fractional part of $\frac{3^{2n}}{8}$, is equal to

(1) $\frac{3}{8}$

(2) $\frac{7}{8}$

(3) $\frac{1}{8}$

(4) None of these

Q7 - Single Correct

If the unit digit of $13^n + 7^n - 3^n, n \in N$, is 7, then the value of n is of the form

(1) $4k+1, k \in 1$

(2) $4k+2, k \in 1$

(3) $4k+3, k \in 1$

(4) $4k, k \in I$

Q8 - Single Correct

The value of $S=\sum_{r=0}^{n-1}\frac{1}{(n-r)^2}\cdot\left(\frac{{}^nC_{r+1}}{{}^nC_r}\right)^2$ equals

(1) $\frac{1}{n+1}$

(2) $\frac{2}{n+1}\left(1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n}\right)$

(3) $\frac{1}{n+1}\left(1+\frac{1}{2}+\cdots+\frac{1}{n}\right)$

(4) $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots+\frac{1}{n^2}$

Q9 - Single Correct

$(x+m)^m - {}^mC_1(x+m-1)^m + {}^mC_2(x+m-2)^m - \cdots + (-1)^m \cdot x^m$ is equal to

(1) $(m+1)!$

(2) $m+1p_m$

(3) $m!$

(4) ${}^mP_m$

Q10 - Single Correct

The number $(2010)^n - 2(1010)^n - 4(510)^n + 5(10)^n, n \in N$ is always divisible by

(1) 10^4

(2) 10^3

(3) 10^5

(4) None of these

Q11 - Multiple Correct

If $(9+\sqrt{80})^n = I + f$, where I, n are integers and $0 < f < 1$, then

(1) I is odd integer

(2) I is an even integer

(3) $(I+f)(1-f) = 1$

(4) $1 - f = (9-\sqrt{80})^n$

Q12 - Multiple Correct

If ${}^{100}C_6+4\cdot{}^{100}C_7+6\cdot{}^{100}C_8+4\cdot{}^{100}C_9+{}^{100}C_{10}$ has the value equal to ${}^{x}C_y$, then the value of $(x+y)$ can be

(1) 114

(2) 115

(3) 198

(4) 199

Q13 - Multiple Correct

The value of the constant term in the binomial $\left(x^2+\frac{1}{x^2}-2\right)^{10}$ is also equal to

(1) number of different dissimilar terms in $(x_1+x_2+\cdots+x_{10})^{10}$

(2) $\left({}^{10}C_0\right)^2+\left({}^{10}C_1\right)^2+\cdots+\left({}^{10}C_{10}\right)^2$

(3) coefficient of x^{10} in $(1-x)^{20}$

(4) number of linear arrangements of 20 things of which 10 alike of one kind and rest all are different, taken all at a times.

Q14 - Multiple Correct

The coefficient of x^{50} in $\sum_{k=0}^{100}{}^{100}C_k(x-2)^{100-k}\cdot 3^k$ is also equal to

(1) number of ways in which 50 identical books can be distributed in 100 students, if each student can get at most one book.

(2) number of ways in which 100 different white balls and 50 identical red balls can be arranged in a circle, if no two red balls are together.

(3) number of dissimilar terms in $(x_1+x_2+x_3+\cdots+x_{50})^{51}$

(4) $\frac{2\cdot6\cdot10\cdot14\cdots198}{(50)!}$

Q15 - Multiple Correct

The value of ${}^{200}\mathrm{C}_0-{}^{200}\mathrm{C}_1+{}^{200}\mathrm{C}_2-{}^{200}\mathrm{C}_3+\cdots+{}^{200}\mathrm{C}_{148}$ is equal to

(1) ${}^{198}\mathrm{C}_{148}$

(2) ${}^{198}C_{51}$

(3) ${}^{200}C_{148}$

(4) ${}^{199}\mathrm{C}_{148}$

Q16 - Multiple Correct

The 4 th term in the expansion of $\left(2+\frac{3x}{8}\right)^{10}$ has the maximum numerical value, then the value of x can be, in the interval.

(1) $\left[\frac{-64}{21},-2\right]$

(2) $\left[2,\frac{64}{21}\right]$

(3) $\left(-\infty,\frac{-64}{21}\right)$

(4) $\left(\frac{64}{21},\infty\right)$

Q17 - Multiple Correct

In the expansion of $(x+y+z)^{10}$ which of the following will be true

(1) The expansion will contains 66 terms

(2) The coefficient of $x^3y^2z^5$ will be $\frac{(10)!}{3!2!5!}$

(3) There cannot be term of $x^3y^2z^4$

(4) $\sum_{\alpha+\beta+\gamma=10}\frac{(10)!}{\alpha!\beta!\gamma!}=3^{10}$

Q18 - Multiple Correct

Which one of the following is true?

(1) If $x^2<1$, the coefficient of x^n in $\frac{(1+2x)^n+(1+3x)^n}{(1-x)}$ is 3^n+4^{n}

(2) The value of $\sum_{r=0}^{n}(-1)^r\cdot{}^nC_r\left(\frac{1}{2^r}+\frac{3^r}{2^{2r}}+\frac{7^r}{2^{3r}}+\cdots\infty\right)$ is equal to $\left(\frac{1}{2^n-1}\right)$.

(3) The coefficient of x^2 in $\left(1+x+x^2\right)^{10}$ is ${}^{12}C_2$.

(4) The coefficient of x^2 in $\left(1+x+x^2\right)^{10}$ is ${}^{11}C_2$.

Q19 - Multiple Correct

If $1+(1+x)+(1+x)^2+\cdots+(1+x)^n=\sum_{k=0}^n a_k x^k$, then

(1) $\sum_{k=0}^n a_k=2^{n+1}-1$

(2) $a_{n-2}=\frac{n(n+1)}{2}$

(3) $a_p>a_{p-1}$ for $p<\frac{n}{2}$

(4) $\left(a_9\right)^2-\left(a_8\right)^2={ }^{n+2} C_{10} \cdot\left({ }^{n+1} C_{10}-{ }^{n+1} C_9\right)$

Q20 - Multiple Correct

If $f(m)=\sum_{i=0}^m{ }^{30} C_{30-i} \cdot{ }^{20} C_{m-i}$, then

(1) maximum value of $f(m)={ }^{50} C_{25}$

(2) $\sum_{r=0}^{50} f(r)=2^{50}$

(3) $f(m)$ is divisible by 50

(4) $\sum_{m=0}^{50}(f(m))^2={ }^{100} C_{50}$

Q21 - Multiple Correct

The value of ${ }^{1000} C_{50}+{ }^{999} C_{49}+{ }^{998} C_{48}+\cdots+{ }^{950} C_0$, is

(1) ${ }^{1001}\mathrm{C}_{50}$

(2) ${ }^{1002} C_{51}-{ }^{1001} C_{51}$

(3) ${ }^{1001}\mathrm{C}_{951}$

(4) ${ }^{1002} C_{51}-{ }^{1001} C_{950}$

Q22 - Multiple Correct

Let $a_n=\left(1+\frac{1}{n}\right)^n$. Then for each $n \in N$

(1) $a_n \geq 2$

(2) $a_n<3$

(3) $a_n<4$

(4) $a_n<2$

Q23 - Multiple Correct

If $(4+\sqrt{15})^n = I + f$, where n is an odd natural number, I is an integer and $0 < f < 1$, then

(1) I is natural number

(2) I is an even integer

(3) $(I+f)(1-f) = 1$

(4) $1 - f = (4-\sqrt{15})^n$

Q24 - Comprehension Type

Passage I (For Question 24, 25)

If $\left(1+px+x^2\right)^n = 1 + a_1x + a_2x^2 + \cdots + a_{2n}x^{2n}$, then

Which of the following is true for $1 < r < 2n$

(1) $(np+pr)a_r = (r+1)a_{r+1} + (r-1)a_{r-1}$

(2) $(np-pr)a_r = (r+1)a_{r+1} + (r-1-2n)a_{r-1}$

(3) $(np-pr)a_r = (r+1)a_{r+1} + (r-1-n)a_{r-1}$

(4) $(2np+pr)a_r = (r+1+n)a_{r+1} + (r+1-n)a_{f-1}$

Q25 - Comprehension Type

The remainder obtained, when $a_1 + 5a_2 + 9a_3 + \cdots + (8n-3)a_{2n}$ is divided by $(p+2)$, is

(1) 1

(2) 2

(3) 3

(4) 0

Q26 - Comprehension Type

Passage II (For Question 26, 27)

If $\left(1+x+x^2\right)^{100} = \sum_{r=0}^{200} a_r x^r$, then

Which of the following is true?

(1) $a_{28} = a_{72}$

(2) $a_{56} = a_{144}$

(3) $a_{200} = a_{300}$

(4) None of these

Q27 - Comprehension Type

$a_0 + a_1 + a_2 + \cdots + a_{99}$, is equal to

(1) $\frac{3^{99} - a_{c19}}{2}$

(2) $\frac{3^{101} - a_{99}}{2}$

(3) $\frac{3^{100} - a_{100}}{2}$

(4) None of these

Q28 - Comprehension Type

Passage III (For Question 28, 29)

Let n be a positive integer such that $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x'$? then

The value: of $\sum \sum_{0 \leq r < s \leq n} (r+s)\left({}^nC_r + {}^nC_s\right)$, is

(1) $n^2 \cdot 2^{n-1}$

(2) $n(n-1)2^{n-2}$

(3) $n^2 \cdot 2^{n-2}$

(4) $n^2 \cdot 2^n$

Q29 - Comprehension Type

The value: of $\sum \sum_{0 \leq r < s \leq n} (r+s)\left({}^nC_r + {}^nC_s + {}^nC_r \cdot {}^nC_s\right)$, is

(1) $n^2 \cdot 2^n \cdot \frac{n}{2}\left[2^{2n} - {}^{2n}C_n\right]$

(2) $n^2 \cdot 2^n + \frac{n}{2}\left[2^{2n} - {}^{2n}C_n\right]$

(3) $n^2 \cdot 2^n - \frac{n}{2}\left[2^{2n} + {}^{2n}C_n\right]$

(4) $n^2 \cdot 2^n + \frac{n}{2}\left[2^n - 1\right]$

Q30 - Comprehension Type

Passage IV (For Question 30, 31)

For $n \in N$, consider $\left(1 + x + x^2\right)^n = \sum_{r=0}^{2n} a_r x^r$, then

If $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \cdots + a_{2n}^2 = k \cdot a_n$ then k is equal to

(1) 1

(2) 2

(3) $\frac{1}{3}$

(4) 0

Q31 - Comprehension Type

If $a = \sum_{k=0}^{[n/2]} {}^nC_{2k} \cdot {}^{2k}C_k$, where $[x]$ denotes greatest integer less than or equals to x, then $a_n - a$ is

(1) 0

(2) 1

(3) -1

(4) 2

Q32 - Comprehension Type

Passage V (For Question 32, 33)

If n is a positive integer and x is any real number, then $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$,

where ${}^nC_0, {}^nC_1, {}^nC_2, \cdots, {}^nC_n$ are binomial coefficients, then

The value of $\left({}^nC_1 - 3 \cdot {}^nC_3 + 5 \cdot {}^nC_5 - \cdots\right)^2 + \left(2 \cdot {}^nC_2 - 4 \cdot {}^nC_4 + 6 \cdot {}^nC_6 - \cdots\right)^2$, is

(1) $n \cdot 2^{n-1}$

(2) $n^2 \cdot 2^{n-1}$

(3) $n(n-1) \cdot 2^{n-1}$

(4) $n^2 \cdot 2^n$

Q33 - Comprehension Type

$\frac{{}^nC_0}{n(n+1)} - \frac{{}^nC_1}{(n+1)(n+2)} + \frac{{}^nC_2}{(n+2)(n+3)} - \cdots$ upto $(n+1)$ terms is equal to

(1) $\int_0^1 x^{n+1}(1-x)^{n+1} dx$

(2) $\int_0^1 x^n(1-x)^{n+1} dx$

(3) $\int_0^1 x^{n-1} \cdot (1-x)^{n+1} dx$

(4) $\int_0^1 x^n(1-x)^{n-1} dx$

Q34 - Integer Type

If $\sum_{k=0}^{100} \left(\frac{k}{k+1}\right)^{100} C_k = \frac{a(2^{100})+b}{c}$, where a, b and $c \in N$, then find the number of digits in least value of $(a+b+c)$

Q35 - Integer Type

The value of $\lim_{n\to\infty} \sum_{r=0}^{n} \frac{{}^nC_r}{n^r(r+3)} = e - x$, then x is

Q36 - Integer Type

$\lim_{x\to 0} \left(\sum_{r=0}^{n} (-1)^r \cdot {}^nC_r \cdot \left(\sum_{k=0}^{n-r} {}^{n-r}C_k \cdot x^k \cdot 2^k\right) \cdot \left(x^2 - x\right)^r\right)^{1/x} = e^{n\lambda}$, then λ is equal to

Q37 - Integer Type

If ${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \cdots + (-I)^{r\,n}C_r = 28$, then n is equal to

Q38 - Integer Type

If $S = \sum_{k=1}^{\infty} \frac{1}{k^2}$ and $t = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2}$, then the value? of $\frac{S}{t}$ is equal to

Q39 - Integer Type

Find the last two digits of 17^{256}.

Q40 - Integer Type

The value of $\frac{y_1 \cdot y_2 \cdot y_3}{501(y_1-x_1)(y_2-x_2)(y_3-x_3)}$ when $(x_i, y_i), i=1,2,3$ satisfy both $x^3-3xy^2=2005 \& y^3-3x^2y=2004$ is

Answer Key

Q1 (3)	**Q2** (1)	**Q3** (4)	**Q4** (3)
Q5 (3)	**Q6** (3)	**Q7** (1)	**Q8** (4)
Q9 (3)	**Q10** (2)	**Q11** (1,3,4)	**Q12** (1,3)
Q13 (2,3)	**Q14** (1,4)	**Q15** (4)	**Q16** (1,2)
Q17 (1,2,3,4)	**Q18** (1,2,4)	**Q19** (1,2,3,4)	**Q20** (1,2,4)
Q21 (1,2,3,4)	**Q22** (1,2,3)	**Q23** (1,3,4)	**Q24** (2)
Q25 (3)	**Q26** (2)	**Q27** (3)	**Q28** (4)
Q29 (2)	**Q30** (1)	**Q31** (2)	**Q32** (2)
Q33 (3)	**Q34** (3)	**Q35** (2)	**Q36** (3)
Q37 (9)	**Q38** (2)	**Q39** (81)	**Q40** (2)